Assignment Deck

Associate Product Manager

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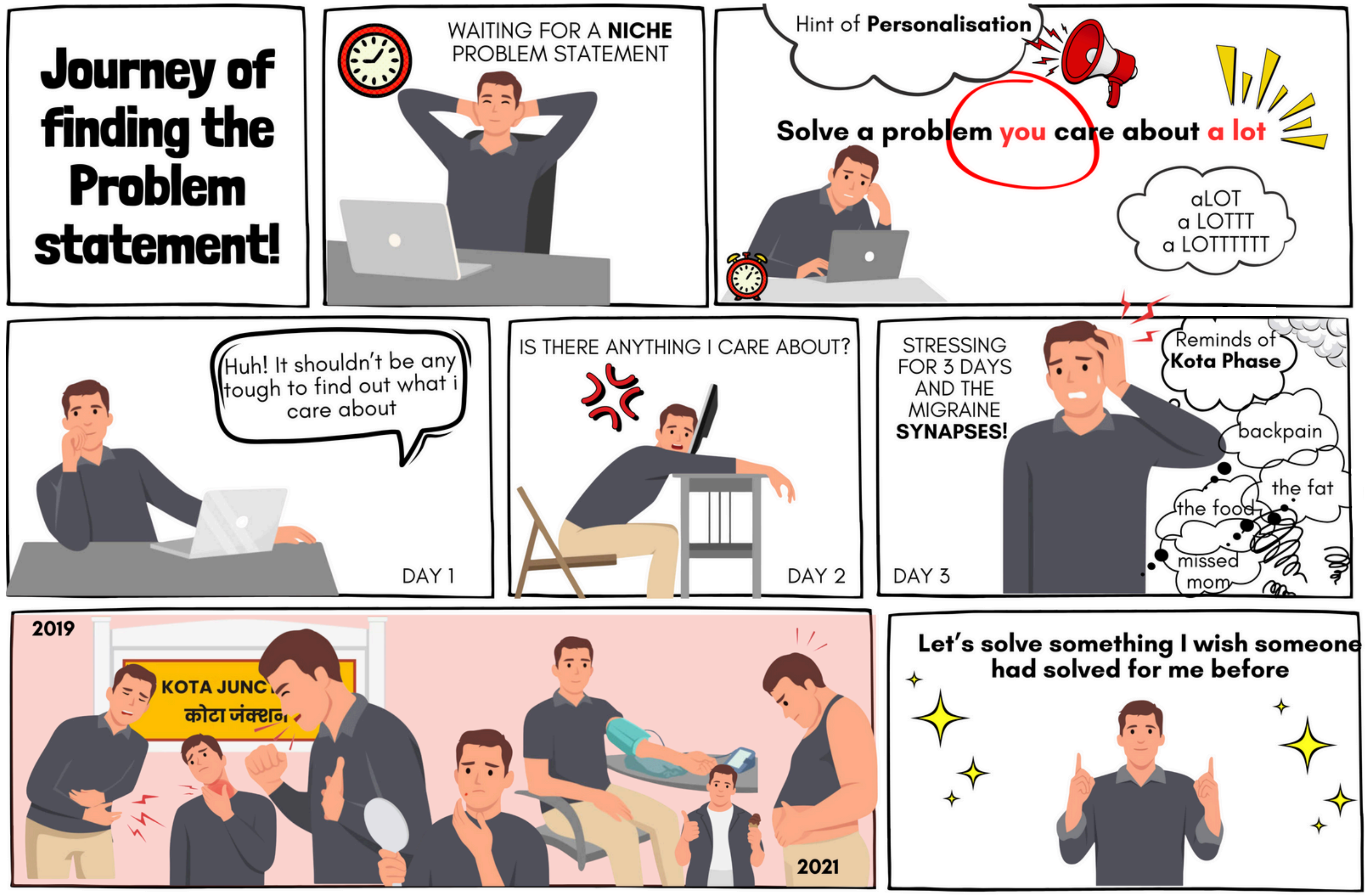
2021BB10362

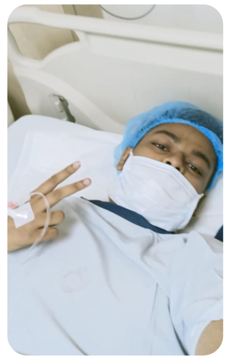


Personal experience

I left home at the **age of 15** to pursue my 'IIT dream'. The ambitions came at a cost of my health. I acquired an infection, a bad posture, put on weight and a chronic disease. Though now, I have worked upon many of them but the chronic disease is going to stay **for lifetime**.

Year	**2019**
Location	**Hometown**
Weight	**67 Kgs**
Surgeries	**None**
Diseases Diagnosed	**None**

2 years →



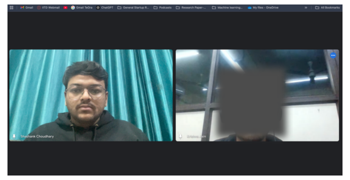
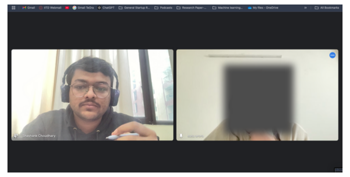
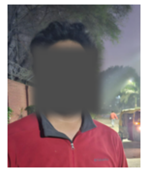
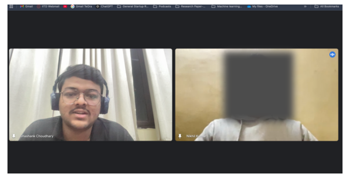
Year	**2021**
Location	**Kota, Raj.**
Weight	**94 Kgs**
Surgeries	**2**
Diseases Diagnosed	**3**

Framing the problem

Preventable diseases that strike early in the lives of students staying away from their parents due to health mismanagement often persist for a lifetime.

What do current students have to say about their lifestyle?

We interviewed 7 students one-on-one across different cities preparing for competitive exams to get a first hand view of their living situation. This helped us conduct further primary research.

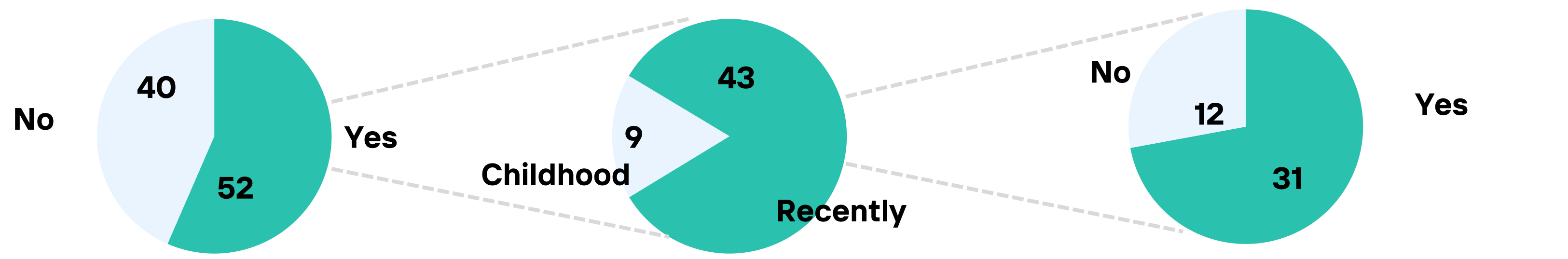
Student	**Name/Age**	**Location/ Exam**	**Complains about**
	Shreyas K., 17	Hyderabad, IIT-JEE	Strict social rules at her coaching makes her feel confined and homesick. Finds it **difficult to sleep** sound at night.
	Suhasa Arora, 18	Kota, Raj. NEET-UG	Over-spicy and subpar **mess food** **Hair fall** due to hard water. Frequent **headaches**.
	Sudhansu S., 21	New Delhi, IIT-JAM	Misses his old college friend. Allergic episodes of **asthma**. Caught **Dengue** once.
	Pankaj Nagar, 20	New Delhi, CA-Level 2	Misses his hometown friends and life. **Back pain** due to long sitting hours. Bathing frequency has been reduced.

Primary Research

We surveyed **92 IIT Delhi recent alumni** through google form and talked to **12 parents** on call to get a deeper understanding of the health mismanagement issue and **break down the problem** further.

What do past students have to say about their health?

Do you have any health ailment currently? → Did you have it since childhood or have you developed it recent years? → Did you neglect the symptoms at first?



Mean age: 22.3

Gender: Male 64% | Female 36%

Current Profession: Corporate 73% | Studying 12% | Others

Did you tell your parents about symptoms? Yes 53% | No 47%

If No, mention your reason

"They worry unnecessarily"

"I thought symptoms would go on their own"

Identifying parents' pain points through survey

12/12 Parents prioritise their children health over their career

8/12 Parents complain that their kids do not take care of health

6/12 Parents mention that their kids do not share about symptoms

4/12 Parents perform this step:

Parents journey navigating through their kid's health

Adieu farewell to kids → Occasionally sends dry fruits and supplements → Ask about their health daily on call → Book blood tests through local centre/online → Find it difficult to navigate between tests results → End up worrying constantly and leaving it to their kids

Empathising with stakeholders

Student

Abhinav Sharma
CLAT Aspirant

Age/Location : 17/ Sikar, Raj.
Technology : Friendly

Pain Points

- Worries about the **loss of time** due to health issues during preparation.
- Stresses about failing behind with **peers** in career/prep.
- Doesn't want to stress parents about **'small' symptoms**.

Needs

- Motivation to **manage health** without affecting studies.
- Wants parents to **worry less** about him.

Parents

Kailash Sharma
F/O Abhinav

Age/Location : 49/ Mathura, UP
Technology : Uses FB, WA, Paytm, IRCTC

Pain Points

- Cannot assess his kid's well being in real and find out if he is **taking care** of himself.
- His son not following their health advices or taking **meds on time**.
- Have to stay their son for long duration if he's ill; **work affected.**

Needs

- Something to access their kids health while sitting **at home.**
- **Holistic information** so that they don't have to worry unnecessarily.

Coaching/College

Dr. Somnath
Admin ABC Coaching

Coaching Location : Sikar, Raj.
Type of exam prep : Law-UG

Pain Points

- Students **dropping out/** performing bad in exam due to serious health issues.
- Parents and **media blaming** coaching for harsh environment.
- Profits dipping due to rise of alternate **online classes**.

Needs

- **Top ranks** in CLAT exam to celebrate.
- **No media negativity** due to suicides/ student's issues.

Early conclusions from breakdown of problem

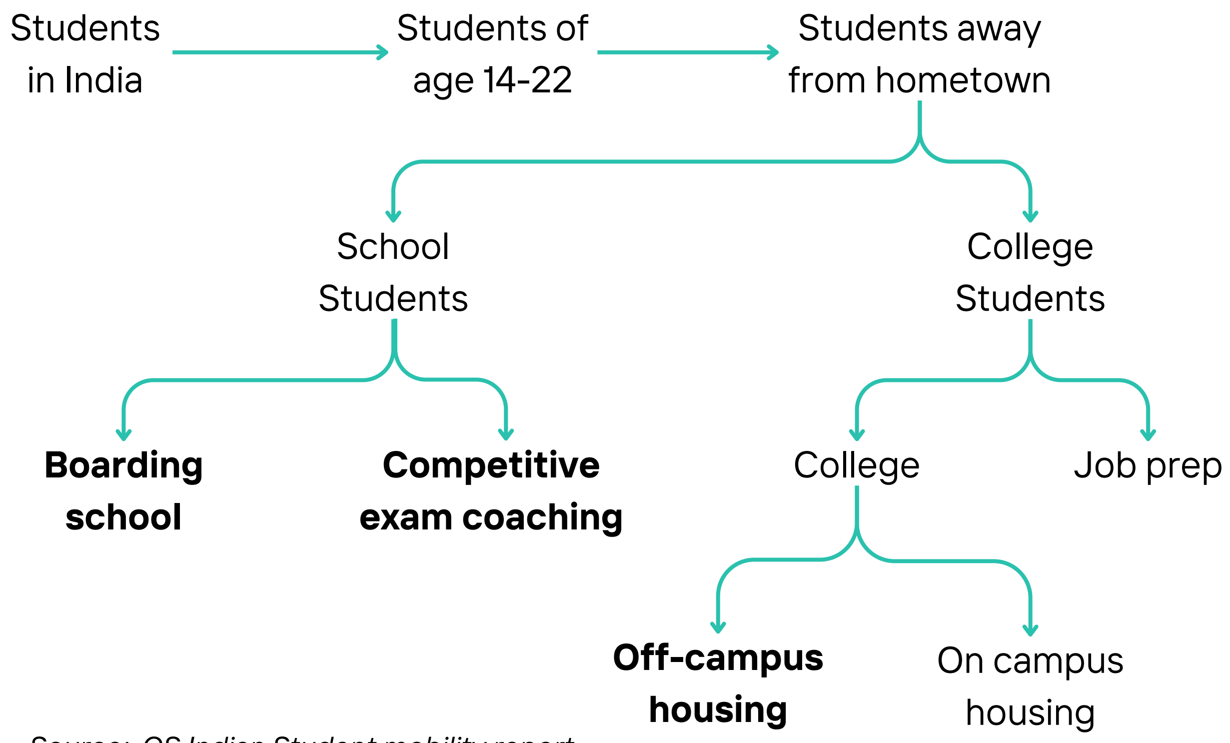
Students prioritizing their careers over health need parental-like healthcare support, at least virtually.

Parents worrying unnecessarily need a robust kid's health monitoring method with holistic information.

Secondary Research

Since the problem has been established, let us understand the who are we solving for first, what is the **clinical basis** to solve this and what **already exists in the market**.

Who should we solve for first and why?

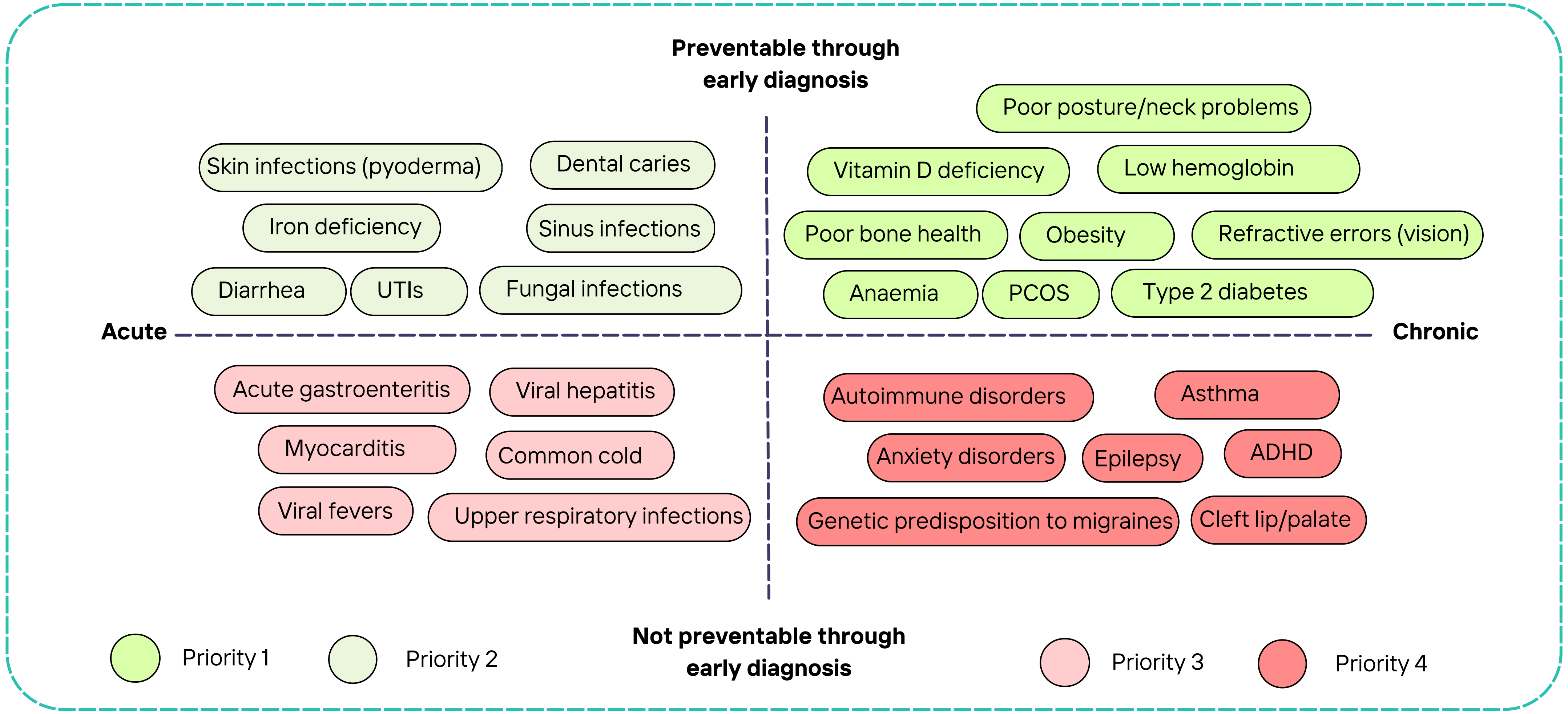


Source: QS Indian Student mobility report

Criteria	Off campus college	Coaching students	Boarding schools
Maturity Level	**Moderate:** Some students have prior independent living experience	**Low:** Most students are living alone for the first time	**Very low:** Most of the students do not have any real world understanding.
Living conditions	**Subpar:** Expensive PGs/flats and reliance on mess food.	**Subpar:** Low quality PGs at inflated rates. Price-minded food messes.	**Good:** Usually well-paid quality messes and expensive hostels.
Existing help	**Some:** Adequate friend circle of to support during crisis.	**Little:** Friends are usually not mature enough to support emotionally.	**Some:** Protected environment with assigned guardians in hostels.

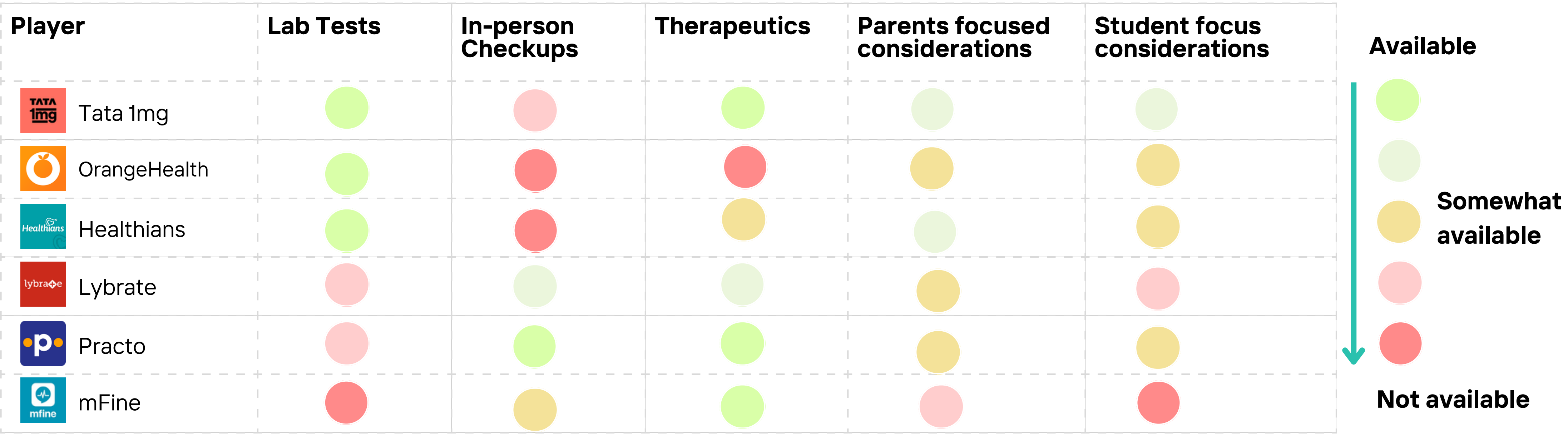
What are the diseases we are talking about and are they preventable?

We have made a list of the most common diseases reported by doctors (especially paediatricians) that occur in young patients of age 14-22 years old and categorised them into a grid on the basis on their type and preventability.



Source: *Praxis OPD Report*

Existing tech solutions in market aiding in student's health management



Source: Self analysis

Solution Flow and Features

Personalised preventive healthcare platform for students, with parents as the users.

An application layer on top of India's vast diagnostics network which will allows parents to book regular and **relevant tests** and checkups, **visualise** the results, **educate** themselves and **analyse** for the next course of action for their kid studying away from home.

Book test

Parents booking test for their child, putting address and **minimal details**

Form Link sent to child's phone

Triage form

Student filling **gamified form** answering about their lifestyle, diet and possible symptoms, if any

Curating personalised test and checkup stack based on form

Clinic day!

Student visiting nearest partner clinic for physical checkup from doctors and blood sample collection. Whole operation time optimised for **60-min checkout**.

Date-time slotting as per student's convenience

Clinical summary

General checkup stats and biomarkers report shared with **designated specialist** for health advice

Early analysis of report to check which specialist to send it to for advice

Report Visualised

Interactive checkup report and knowledge capsules on parents app

Preparing video library for education

Breaking down feedback & findings

Plan of action

Analysis of doctors advice by **breaking down** in clinical and non-clinical advices

Curating clinically relevant tests stack through **personalised** preventive biomarkers and checkups selection automation

Mind and Development

Pain Point

Crucial age for brain and body development. Exposure to stress due to peer pressure.

Biomarkers

- Omega-3: EPA+DPA+DHA
- Cortisol

Eye & Dental

Checkup

- Refraction Assessment
- Routine dental checkup

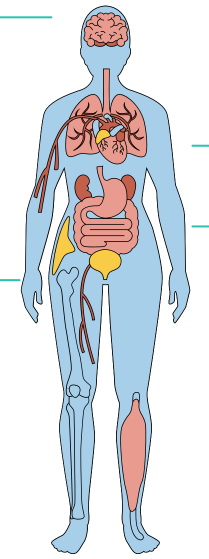
Posture and Hygiene

Pain Point

Long sitting hours. No active lifestyle. Unawareness about hygiene and skincare.

Checkup

- Spinal curvature (kyphosis, scoliosis)
- Plumb Line Test
- Head and neck alignment. Neck hump.
- Screening for skin infections and scabies
- C-reactive proteins (hs-CRP)



Women Health

Pain Point

Lack of mother for personal guidance can lead to doubts. Late diagnosis of PCOS.

Biomarkers

- Follicle Stimulating Hormone (FSH) (female)
- Testosterone, Free (female)

Metabolism

Checkup & Biomarkers

- BMI (Obesity)
- Blood Sugar & CBC
- Blood pressure
- Comprehensive Metabolic Panel (CMP)

Nutritional Deficiencies

Pain Point

Irregular dietary habits, consumption of junk and unawareness about nutrient rich diet

Biomarkers

- Calcium
- Zinc
- Iron Binding Capacity
- Magnesium
- Homocysteine
- Vitamin D

How can we personalise the checkup through automation?

Pain Point

Waste of medical resources and time on unnecessary tests as well as parent's money.

Solution

Developing own NLP for prioritisation on basis of the risk analysis: personalised form response, family history, geography (epidemiology) and previous test reports.

Technical Workflow

Inputs	*Weights*
Student's Lifestyle data through form	Risk analysis based on lifestyle gaps
Family history and previous test reports	Remove already diagnosed & add risks
Current location	Risk of epidemiology based diseases

Activation function

$f(x)$

Output

Prioritised list of biomarkers

Personalised test stack

Designing an experience for parents to **access** their child's health

User Journey

Student's Web App

Filling form sent by parents for triaging and personalised test stack

Parents' App

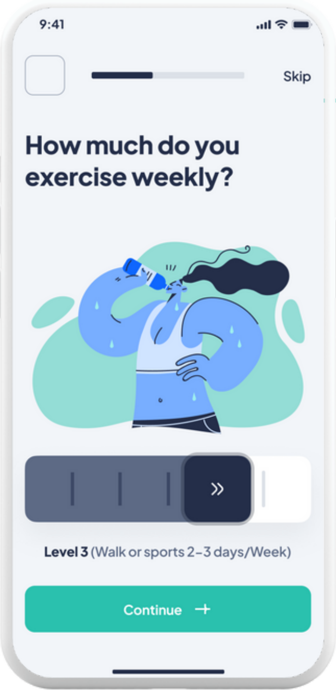
Report Visualisation

Knowledge capsules

Non-clinical follow-up advices

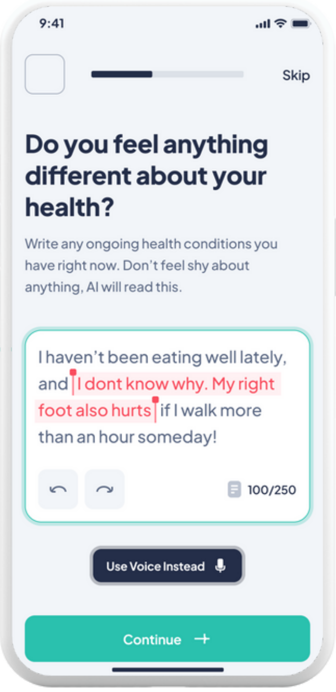
Clinical call-to-action

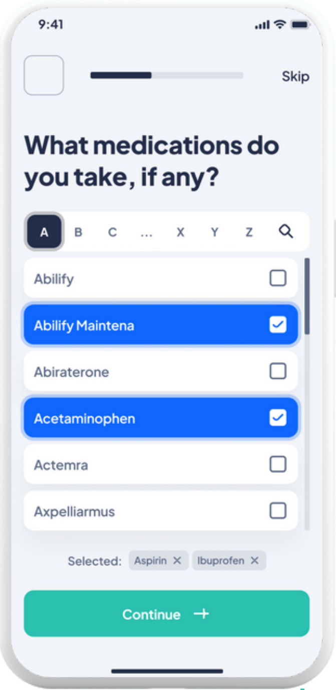
Rough screens and elements



Basic Lifestyle Questions

- Simple & sequential form layout to prevent burnout and fall throughs
- Similar questions on sleep, diet mood, activity, stress & hobbies.





Subjective health description

- Mimics comfortable family doctor with whom you can share anything freely.
- Option to use voice based input for more description and depth.

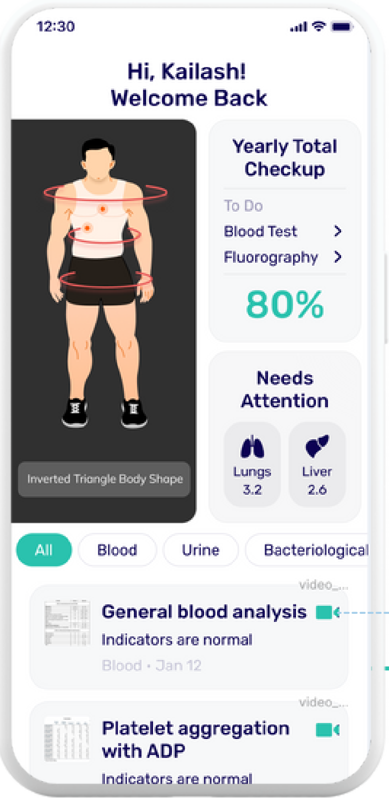
Direct Medical questions

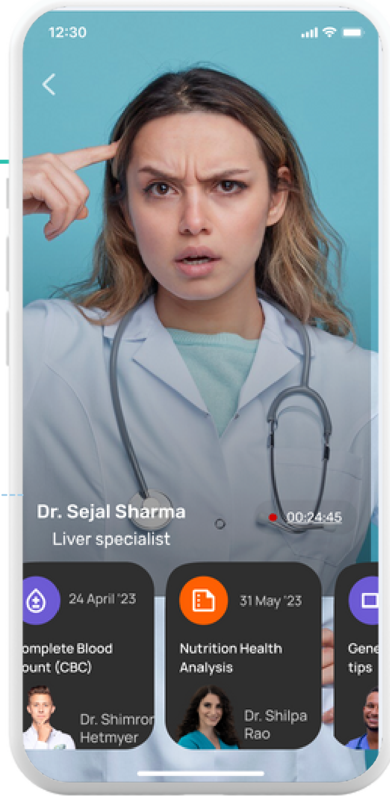
- Questions about family health history, previously diagnosed conditions, medicine taken, vaccination etc

60-sec knowledge capsules for elements of report

Pre-recorded videos on relevant biomarkes

Dashboard



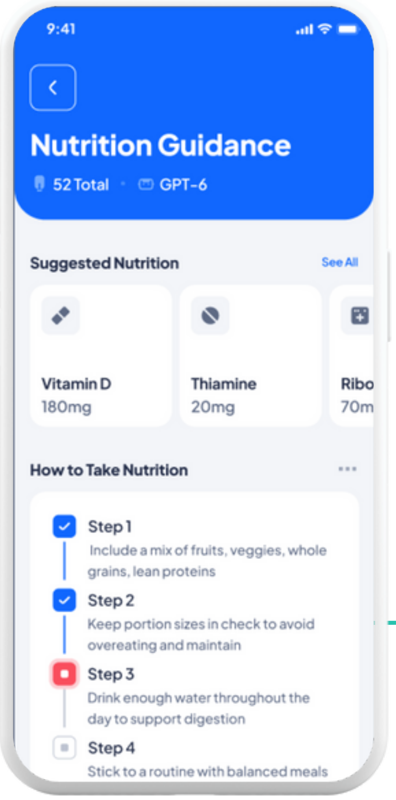


60-sec timer to keep the videos engaging

Content from verified specialist doctors only

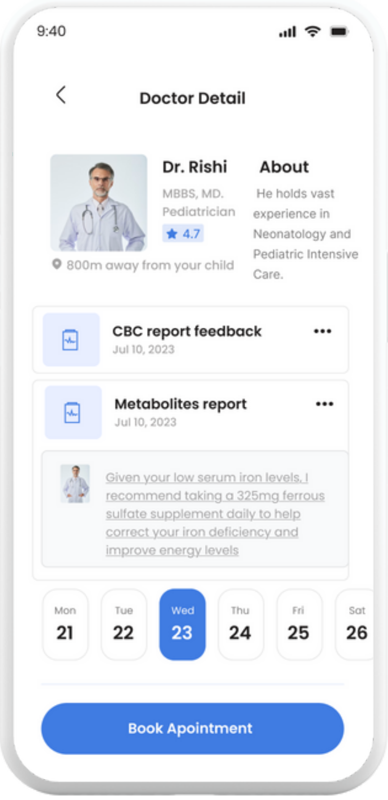
Quick peek into top report insights

- Simple & sequential form layout to prevent burnout and fall throughs



Clinical Feedback and CTA

Specialist take on report and CTA: either an appointment or a follow up test



Non-clinical diet, sleep, yoga etc advices

These are the AI generated advices trained by dieticians and trainers

Pain Points addressed

Long boring forms and consultation procedures build barriers for students who prioritise time

Children **reluctant to share** personal details with doctor. Often, hiding away symptoms

Exclusion of previous EHRs in curating tests makes the diagnostics more expensive and time consuming

Difficulties to navigate within pdf reports. Cannot prioritise biomarkers on basis on 'urgency to respond'.

Consuming untrusted videos from youtube or searching on google about biomarkers. Either end get anxious or ignoring something urgent

Unaware about the structured approaches to follow **home remedies, diet and exercises**. Often miss out the potential of these.

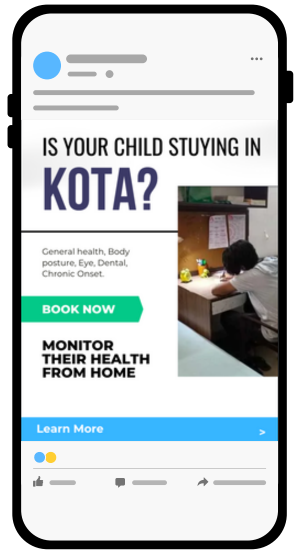
Post-tests consultation drop-offs. Only a few percentage of people actually consult specialists to discuss their report findings.

Product Development

Only if CTR>10% and WLR>2%

MVP 1: 'Fake ads' to analyse click through rate before product development

Objective Demand Forecasting



Instagram ad targeting the focus user group's pain point

Potential user who resonate with the problem might click the link at least once

Coming soon page with an early sign up form for waitlist

We only expect a very few percentage people to fill the waitlist

Metrics

Click-Through Rate (CTR)

What

(Clicks / Impressions) x 100

Expected range

5-10%

Conversion Rate (WLR)

What

(Waitlist signups / Clicks) x 100

Expected range

1-2%

Pre-development

Early development

MVP 2 : Free Health camp at coaching centres, but results can only be found on our app

Objective Product Validation

Type Eye, Dental & Posture examination

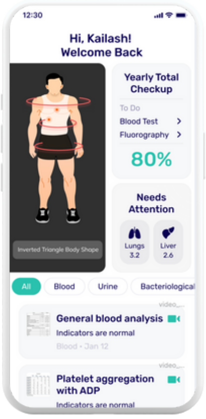
Location Any coaching centre, Kota

Incentive for coaching Positive media statement

Flow

Checkup and preliminary questionnaire along with contact information

Using parent's contact info



Report visualised along with the other features

Metrics

Cold call feedback

Calling parents who have accessed report to know their first hand experience and list problems

In-app Metrics

Feature Discovery Rate

Percentage of users who find or access the new feature.

Drop-Off Points

Locations in the user flow where users disengage

Feature Adoption Rate

The percentage of active users who use the new feature

Repeat Usage Rate

Percentage of users who return to use the feature more than once.

Technical Infrastructure

Technical design goals:

Result accuracy

Anomaly detection

User's data privacy

Use of emerging technology for speed and accuracy

Personalised form for student

EHR in system, if available

Inputs

Data

Biomarkers and tests curation NLP

Dynamic Price

Payment confirmation

Student's details, slot, list of tests/checkups to be conducted

Partner clinic ERP/Dashboard

Checkups' results and physician's suggestions

Proprietary analytics engine

Raw blood test data+ Pathologist's comments

Partner Diagnostic Chain ERP

Abnormal Results

Clinical feedback engine

Specialist 3 working app

Clinical Feedback

Clinical Feedback

Specialist 1 working app

Specialist 2 working app

Normal Results

In-house scrutiny/ anomaly detection

Final test results for education content curation

Library of informative videos, articles and structured guidance

Final report

Data visulisation

Visualised results

Parents App

User feedback analytics system

Regulatory compliances

Compliance with SaMD guidelines

SaMD stands for software as a medical devices. **FDA (for US) and CDSCO (for India)** have guidelines for AI applications in healthcare.

FDA's Good ML practices :

- Clinical Study Participants and Data Sets Are Representative of the Intended Patient Population.
- Training Data Sets Are Independent of Test Sets.
- Effective transparency: ensures that information that could impact risks and patient outcomes is communicated.

Data protection, Privacy and permissions

Ensuring patient data is handled securely and in accordance with data privacy regulations: **HIPAA (US) & DISHA (India)**

Consideration in our app for 18+ students:

Parents send the triage form link to their child

Student digitally agrees to share his personal medical data (report) with his/her parents

YES
Parents get access to the reports on their app

NO
Report sent only to student

Design considerations

Focus Group

Parents (Age 45-60)

Tier 2,3 residents

Familiar with Paytm, IRCTC, Facebook

Color palette

General App		Positivity, calm, wellness
Clinical pages		Trust & Transparency
Texts: Dark		Legible in small fonts too

Special considerations

Feature	Inclusions
Design for Touch	Larger Touch Targets, Familiar gestures only
Enhance Accessibility	Voice Assistance, Haptic Feedback
Build Trust	Privacy Assurance, No Overwhelming Permissions
Reduce Cognitive Load	Familiar Patterns, Pre-fill or suggest defaults
Simplify Navigation	Clear Menus, Fewer Steps, Progress indicators

Product Success

The goal is to have a platform with mass adoption of accurate student health testing and effective education to parents

Metrics

North Start User Metric: **Value-Driven Engagement Score**

North Star Clinical Metric: **Abnormal Result Detection Rate**

Phase	MVP	Development	Early Traction	Retention	Scaling
Focus	Demand and concept Validation	Improving Usability and features	User Acquisition and Engagement	Fostering long-term relationships	Expanding Product Base
User Metrics	"Fake Ad" CTR, MVP usability	DAU/MAU, Feature adoption rate, Load times, Customer Support tickets	App downloads, Referral rate	Repeated bookings, Churn rate	Revenue per user (ARPU), Lifetime value (LTV), LTV/CAC ratio
Clinical Metrics	Report turnaround time	NLP accuracy in test recommendations, Specialist feedback time	Abnormal result detection rate	Chronic condition early detection rate	Test result consistency across regions

Important User metrics

Value-Driven Engagement Score

Monthly active users / Total Users) × (Feature Usage Rate)

Reflects the goal of educating parents along with the tests

Integrated Booking Efficiency

Booking Efficiency = (Tests Booked / App Visits) × (Test Accuracy / Report Turn around Time) × 100

Overall effectiveness of the app in convertions and quality of the process.

Feature Impact Score

(DAU / MAU) × Feature Adoption Rate × (1 / App Load Time)

Combines quantified user adoption while considering app's performance

Important Clinical metrics

Abnormal Result Detection Rate

(Number of Abnormal Results Correctly Identified) / (Total Number of Tests Conducted)

Reflects its clinical value and ability to support early detection

NLP accuracy in test recommendations

F1-score = 2 × (Precision × Recall) / (Precision + Recall)

NLP's effectiveness in correctly identifying and suggesting relevant health tests for user

Total Response Efficiency (TRE)

TRE = 1 / (Report Turnaround Time + Specialist Feedback Time)

Provides a comprehensive view of the system's responsiveness

Pitfalls and mitigation

Decreasing TAM due to increase in online education

Arguement This growth will be backed by a phenomenal rise in the paid user base for online education in India, which is expected to grow from the current base of 1.57 million users to 9.5 million users in 2021 at a **CAGR of 44%**.

Counter Migration from hometown to metros is **not zero sum**. While people who could not afford to shift to expensive coaching centres are adopting online, a new market is getting created for students who earlier used to join local colleges shifting to different cities (or even countries) for better college education.

Mitigation **Constant evolution** into adjacent markets as the time changes. Currently coachings are the prime focus. It can be further expanded into college students or students abroad.

Cost of training own NLP might not be justified in cost-benefit analysis

Arguement The cost associated with datasets and training to train near perfect models in high and can only in afforded by big tech companies. **Cost of FDA approval** is an add-on

Mitigation **IP** should be protected to ensure investors' interest. Development plans should be made in **phase-wise manner** only when a large scale and demand is visible.

Operational & distribution upper-hand of diagnostics chain competitors

Arguement Chains like TATA 1mg, Pharmeasy, Healthians, Orangehealth who might not focus on student market now **can easily expand** due to their existing distribution.

Mitigation **Partner** with the large offline network and the 'competitor' chains by leveraging our **technical (NLP) capability** and **niche expertise** of parent-child segment.

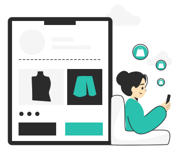
Overestimation of parents' technical know-how

Mitigation User experience for parents should be kept a priority in design. **Offline channels** for booking tests can later be explored to by-pass this issue, especially in tier-3 cities.

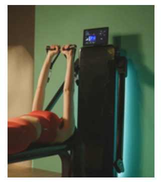
Future expansion possibilities

Focusing on the youth of a country with **50% of population under the age of 25** gives us an advantage to expand omnichannel and omnipresent >7 years down the line.

The current healthcare regime poses multiple challenges which can be solved through **digital health.** Application of AI and other novel technologies are fuelling this process.



Hardware products focused on student health can be sold through our platform



Compact home-gym equipments

Students do minimal outdoor sports and workouts

Lumbar support extension for chair

Often feel physical pain in back due to sitting for long hours

Ready to eat healthy snacks

Do not have access to quality healthy snacks

Digital therapeutics for student's mental health challenges

Peer groups for students with similar age group for healthy weekly interaction to cure loneliness

Cognitive app-based meditation exercises and games for mild anxiety episode

'Reverse Model': Grown up children taking care of their parents' health through same platform

Our users (students) now will be the earning individuals in next 5-7 years

They might have already witnessed the clinical value addition of preventive tests

\+

Our platform's trust and brand awareness across the years would have increased

=

Preventive health platform for elder population with their children as the payers